

Oral Radiograph Anatomical Landmarks

The objectives of this information are to qualify students to read and interpret the normal landmarks visible on intraoral films and to differentiate between the variable structures within the oral cavity.

A strong knowledge of the location and normal appearance of these landmarks is crucial for identifying and orienting radiographs. This knowledge is valuable for dentists in determining whether an area is normal or abnormal.

Basic Radiographic Concepts

Radiolucent vs. Radiopaque

- **Radiolucent (Black):** Structures that are cavities, depressions, or openings in bone, such as a sinus, fossa, canal, or foramen, allow x-rays to penetrate them. These areas appear radiolucent or black on radiographic images.
- **Radiopaque (White):** Structures that are bony in origin will absorb or stop the penetration of x-rays. These areas appear radiopaque or white. Some structures, which partially absorb radiation, are represented in varying degrees of radiopacity.

Types of Bone

- **Cortical Bone:** Also referred to as compact bone, this is the dense outer layer of bone.
 - **Cancellous Bone:** Also called trabecular bone, this is the soft, spongy bone found between two layers of dense cortical bone and is composed of trabeculae.
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anatomical Structures in Radiographs

Normal Tooth Anatomy

The following tooth structures can be viewed on dental images:

- **Enamel:** This is the densest structure found in the human body. It is the outermost radiopaque layer of the crown.
- **Dentin:** Found beneath the enamel, the dentin surrounds the pulp cavity. It appears radiopaque and makes up the majority of the tooth structure, though it is not as radiopaque as enamel.
- **Pulp Cavity:** This consists of a pulp chamber and pulp canals. It contains blood vessels, nerves, and lymphatics and appears relatively radiolucent on a dental image.

- **Cementum:** This structure is not usually apparent radiographically because the cementum layer is very thin.

Supporting Structures

The alveolar process, or alveolar bone, serves as the supporting structure for the teeth. Its anatomical landmarks include the lamina dura, the alveolar crest, and the periodontal ligament space.

- **Lamina Dura:**
 - **Description:** The wall of the tooth socket that surrounds the root of a tooth. It is made up of dense cortical bone.
 - **Appearance:** On a dental image, it appears as a dense radiopaque line surrounding the root.
 - **Alveolar Crest:**
 - **Description:** The most coronal portion of alveolar bone found between teeth. It is made of dense cortical bone and is continuous with the lamina dura.
 - **Appearance:** It appears radiopaque and is typically located 1.5 to 2.0 mm below the cemento-enamel junction (the junction of the crown and root).
 - **Periodontal Ligament (PDL) Space:**
 - **Description:** The space found between the root of the tooth and the lamina dura. This space contains connective tissue fibers, blood vessels, and lymphatics.
 - **Appearance:** It appears as a thin radiolucent line around the tooth's root. In a healthy periodontium, this space is a continuous radiolucent line of uniform thickness.
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General Radiographic Terms

Prominences of Bone

These are composed of dense cortical bone and appear radiopaque on dental images.

- **Process:** A marked prominence or projection of bone (e.g., coronoid process).
- **Ridge:** A linear prominence or projection of bone (e.g., external oblique ridge).
- **Spine:** A sharp, thorn-like projection of bone (e.g., anterior nasal spine).
- **Tuberosity:** A rounded prominence of bone (e.g., maxillary tuberosity).

Spaces and Depressions in Bone

These areas do not resist the passage of x-rays and appear radiolucent.

- **Canal:** A tube-like passageway through bone containing nerves and blood vessels (e.g., mandibular canal).
- **Foramen:** An opening or hole in bone that permits the passage of nerves and blood vessels (e.g., mental foramen).

- **Fossa:** A broad, shallow, scooped-out or depressed area of bone (e.g., submandibular fossa).
- **Sinus:** A hollow space, cavity, or recess in bone (e.g., maxillary sinus).

Miscellaneous Terms

- **Septum:** A bony wall or partition that divides two spaces or cavities (e.g., nasal septum).
 - **Suture:** An immovable joint that represents a line of union between adjoining bones of the skull (e.g., median palatine suture).
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Bony Landmarks of the Maxilla

- **Incisive Foramen (Nasopalatine Foramen):**
 - **Description:** An opening in bone at the midline of the anterior portion of the hard palate.
 - **Appearance:** On an anterior maxillary periapical image, it appears as a small, ovoid or round radiolucent area located between the roots of the maxillary central incisors.
- **Median Palatal Suture:**
 - **Description:** The immovable joint between the two palatine processes of the maxilla.
 - **Appearance:** Appears as a thin radiolucent line between the maxillary central incisors.
- **Lateral Fossa (Canine Fossa):**
 - **Description:** A smooth, depressed area of the maxilla located between the maxillary canine and lateral incisors.
 - **Appearance:** Appears as a radiolucent area between the maxillary canine and lateral incisor.
- **Nasal Cavity (Nasal Fossa):**
 - **Description:** A pear-shaped compartment of bone located superior to the maxilla.
 - **Appearance:** Appears as a large, radiolucent area superior to the maxillary incisors.
- **Nasal Septum:**
 - **Description:** A vertical bony wall or partition that divides the nasal cavity.
 - **Appearance:** Appears as a vertical radiopaque partition that divides the nasal cavity.
- **Floor of Nasal Cavity:**
 - **Description:** The bony wall forming the floor of the nasal cavity.
 - **Appearance:** Appears as a dense radiopaque band of bone superior to the maxillary incisors.
- **Anterior Nasal Spine:**
 - **Description:** A sharp projection of the maxilla at the anterior and inferior portion of the nasal cavity.
 - **Appearance:** Appears as a V-shaped radiopaque area at the intersection of the floor of the nasal cavity and the nasal septum.
- **Maxillary Sinus:**
 - **Description:** Paired cavities or compartments of bone located within the maxilla.
 - **Appearance:** On posterior maxillary periapical images, it appears as a radiolucent area superior to the apices of maxillary premolars and molars.
- **Septa Within Maxillary Sinus:**

- **Description:** Bony walls or partitions that appear to divide the maxillary sinus into compartments.
- **Appearance:** They appear as radiopaque lines within the maxillary sinus.
- **Inverted Y:**
 - **Description:** Refers to the intersection of the maxillary sinus and the nasal cavity.
 - **Appearance:** On a maxillary canine periapical image, it appears as a radiopaque upside-down Y formed by the intersection of the lateral wall of the nasal fossa and the anterior border of the maxillary sinus.
- **Maxillary Tuberosity:**
 - **Description:** A rounded prominence of bone that extends posterior to the third molar region.
 - **Appearance:** Appears as a radiopaque bulge distal to the third molar region.
- **Zygomatic Process of Maxilla:**
 - **Description:** A bony projection of the maxilla, composed of dense cortical bone, that articulates with the zygoma (malar bone).
 - **Appearance:** On a posterior maxillary periapical image, it appears as a J-shaped or U-shaped radiopacity located superior to the maxillary first molar region.

mandible Bony Landmarks of the Mandible

- **Genial Tubercles**
 - **Description:** Tiny bumps of bone on the internal surface of the mandible that serve as attachment sites for the genioglossus and geniohyoid muscles.
 - **Appearance:** A *ring-shaped radiopacity* located inferior to the apices (roots) of the mandibular incisors.
- **Lingual Foramen**
 - **Description:** A tiny opening or hole in the bone on the internal surface of the mandible, located near the midline and surrounded by the genial tubercles.
 - **Appearance:** A *small, radiolucent dot* inferior to the apices of the mandibular incisors.
- **Mental Ridge**
 - **Description:** A linear prominence of cortical bone found on the *external* surface of the anterior portion of the mandible.
 - **Appearance:** A *thick radiopaque band* that extends from the premolar region to the incisor region.
- **Mental Fossa**
 - **Description:** A scooped-out, depressed area of bone located on the *external* surface of the anterior mandible.
 - **Appearance:** A *radiolucent area* seen *above* the mental ridge on a mandibular periapical image.
- **Mental Foramen**
 - **Description:** An opening on the *external* surface of the mandible, typically in the premolar region. Blood vessels and nerves that supply the lower lip exit through this opening.
 - **Appearance:** A *small, ovoid or round radiolucent area* located in the apical region of the mandibular premolars.
- **Mandibular Canal**

- **Description:** A tube-like passageway that travels the length of the mandible, housing the inferior alveolar nerve and blood vessels.
 - **Appearance:** A *radiolucent band* outlined by two thin *radiopaque lines* (which are the cortical walls of the canal). It is typically seen below or superimposed over the apices of the mandibular molar teeth.
 - **Mylohyoid Ridge (Internal Oblique Ridge)**
 - **Description:** A linear prominence of bone on the *internal* surface of the mandible, serving as an attachment site for the mylohyoid muscle.
 - **Appearance:** A *dense radiopaque band* that extends downward and forward from the third molar region, at the level of the posterior teeth apices.
 - **External Oblique Ridge (External Oblique Line)**
 - **Description:** A linear prominence of bone on the *external* surface of the body of the mandible.
 - **Appearance:** A *radiopaque band* extending downward and forward from the anterior border of the ramus of the mandible.
 - **Submandibular Fossa**
 - **Description:** A depressed, scooped-out area of bone on the *internal* surface of the mandible, located *inferior* to the mylohyoid ridge. The submandibular salivary gland is located in this fossa.
 - **Appearance:** A *radiolucent area* in the molar region, below the mylohyoid ridge.
 - **Inferior border of mandible**
 - **Description:** The lowermost part of the mandible.
 - **Appearance:** Appears as a *dense, broad radiopaque band* of bone.
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Radiographic Appearance of Restorative Materials

- **Radiopaque Restorative Materials (Appear White):**
 - Includes Gold, Silver amalgam, Zinc oxide-eugenol, Zinc phosphate cement, Gutta-percha, Silver points, Metal bands & crowns, Metal wires, and dental implants.
- **Radiolucent Restorative Materials (Appear Dark):**
 - Includes Acrylic, Calcium hydroxide pastes (thin or thick !!!), and Porcelain.